

TELECONNECTIONS

BASIC THEORY: ROSSBY WAVES, RAY TRACING AND STATIONARY WAVES

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Beijing, September 2019

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Lecture 1

The Basics

- Teleconnections – what are they?
- Patterns of Variability
- Rossby Waves, Stationary Waves, Ray Tracing

Lecture 2

Some Current Research

- Numerical experiments.
- Teleconnections from the Tropics
- The North Atlantic Oscillation and Annular Modes

What are they?

- Teleconnections are remote connections between different parts of the world that affect the climate and weather, typically on weekly to seasonal timescales.
- Wikipedia says:
“Teleconnection in atmospheric science refers to climate anomalies being related to each other at large distances (typically thousands of kilometers). The most emblematic teleconnection is that linking sea-level pressure at Tahiti and Darwin, Australia, which defines the Southern Oscillation.”

Two General Kinds

1. Large-Scale Oscillations and Patterns (e.g., ‘see-saws’)
 - El Niño and Southern Oscillation
 - Northern Annular Mode (NAM), SAM, NAO, Southern Oscillation, Pacific Decadal Oscillation, etc,
2. Remote Influences of Localized Events
 - Tropical SST Anomalies influencing North America and Europe
 - El Niño influencing monsoons.
 - etc

Applications and Examples

- Sea surface temperature anomalies and weather forecasting. Sea-surface temperature anomalies may last weeks to months. If they affect the weather remotely it gives us extra predictability.
- Atmospheric Oscillations: NAO, PDO, NAM, SAM are teleconnections. El Niño is a teleconnection (Tahiti to Darwin pressure anti-correlation).
- Do blocking events have teleconnections? Teleconnections *from* monsoons?
- Stratosphere tends to have long timescales. Tropospheric influence? A 'stratospheric bridge'?

A Few Early Papers up to 1980s

Gilbert Walker, 1923: Correlation in Seasonal Variations of Weather, VIII: A Preliminary Study of World Weather. *Memoirs of Indian Meteorological Dept.*

Jerome Namias, 1950s. Seasonal weather forecaster for NOAA. Published books of teleconnections for making long-term weather predictions.

Jacob Bjerknes, 1969: Atmospheric Teleconnections from the Equatorial Pacific. *Mon. Wea. Review.*

Wallace and Gutzler, 1981: Teleconnections in the geopotential height field during the Northern Hemisphere winter. *Mon. Wea. Review*

Hoskins and Karoly, 1981: The steady linear response of a spherical atmosphere to thermal and orographic forcing. *J. Atmos. Sci.*

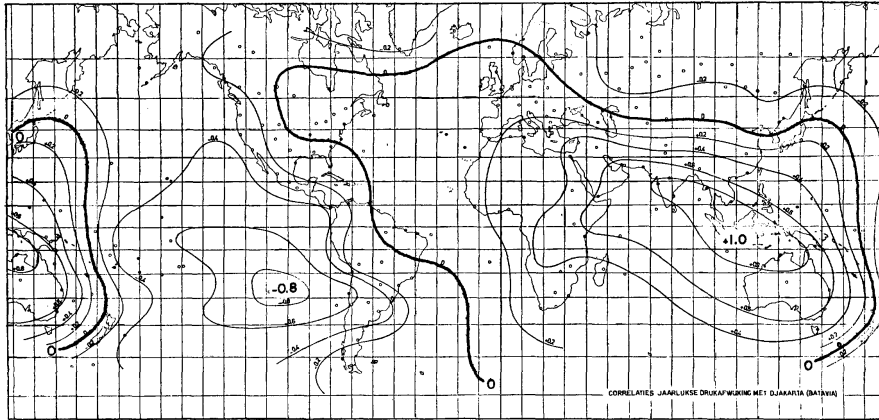


FIGURE 10.—Walker's Southern Oscillation as presented by Berlage (1957). The map shows the worldwide distribution of correlations of annual pressure anomalies with simultaneous pressure anomalies in Djakarta, Indonesia.

Pressure Correlations with Djarkata, Indonesia.

Berlage (1957) and Bjerknes (1969).

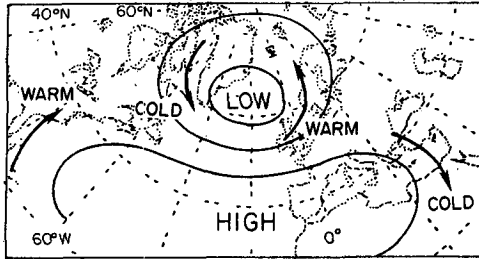


FIG. 1. Idealized relationships between pressure and temperature anomalies associated with the North Atlantic Oscillation.

Wallace and Gutzler, 1981

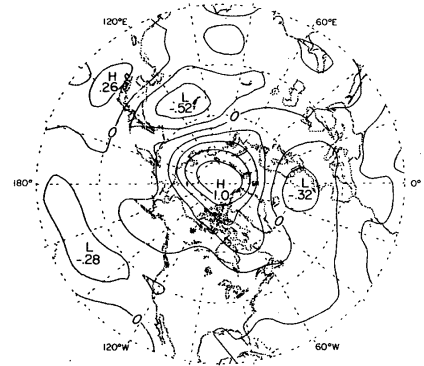
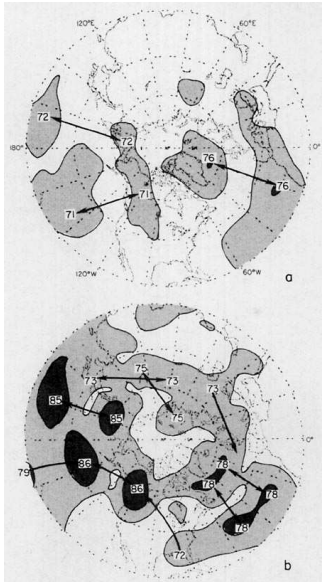


FIG. 6. One-point correlation map showing the correlation coefficient between 500 mb height at the North Pole grid point and 500 mb height every grid point. Based on 45 winter months (Decembers, Januarys and Februarys for the winters 1962–63 to 1976–77). Contour interval is 0.2.

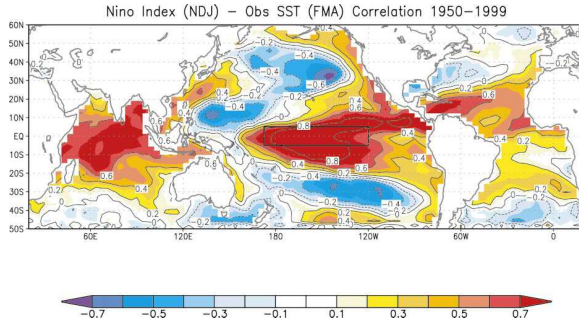


Strongest negative correlations
(in two datasets, 1962–1977 winter)

Sea-level pressure

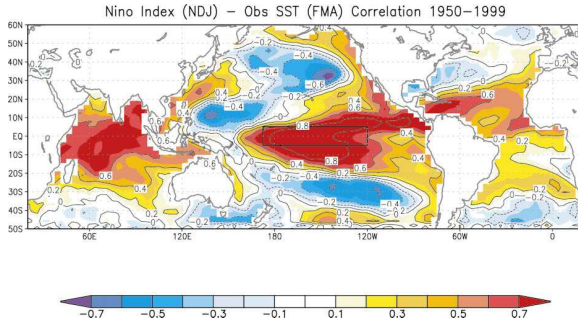
500 hPa height.

Wallace & Gutzler, 1981.

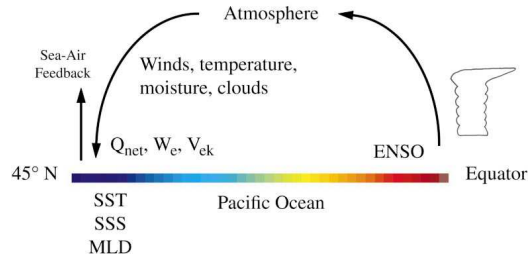


Correlation between El Niño SST anomalies and Global SST anomalies, 1950–1999.

Alexander et al, 2002.



The 'Stratospheric Bridge'



Correlation between El Niño SST anomalies and Global SST anomalies, 1950-1999.

Alexander et al, 2002.

SENSITIVITY OF N. PACIFIC HEIGHT ANOMALIES TO SST

Reverse Engineer the SST Influence

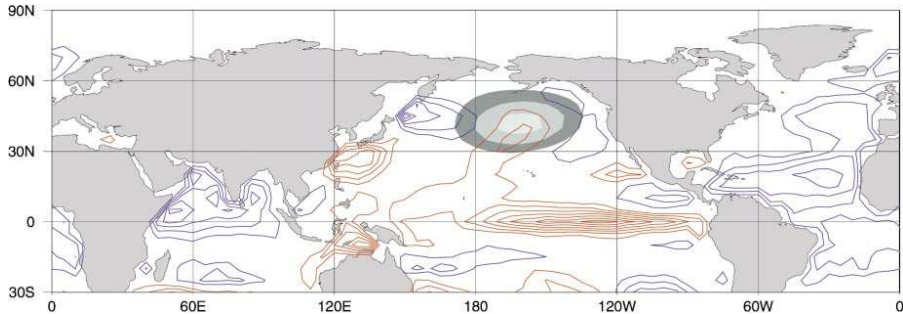
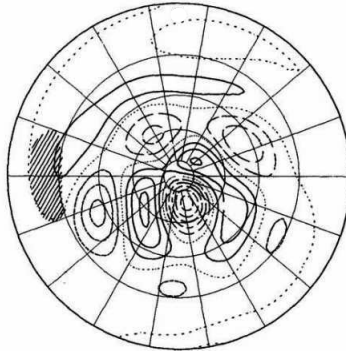
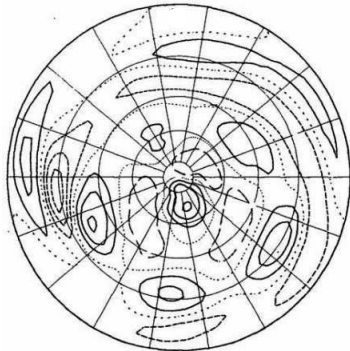
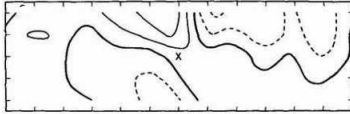


FIG. 15. Sensitivity map, or influence function, of a negative height anomaly over the North Pacific to anomalous SSTs 60 days earlier in the MLM. The influence function is contoured at an interval of 0.002 K m^{-1} ; the zero contour is removed for clarity and positive (negative) values are red (blue). The target height response at 200 mb is indicated by gray shading with a 0.25-m interval.

Influence function, or sensitivity map, of North Pacific height anomalies (grey shading) to SST anomalies (contours) 60 days earlier.

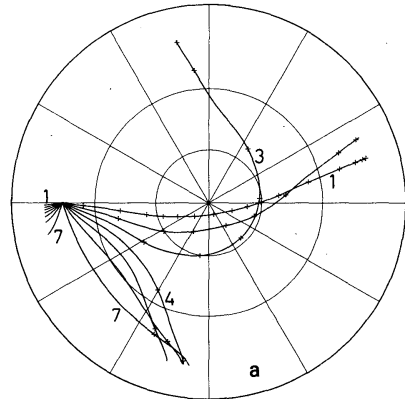
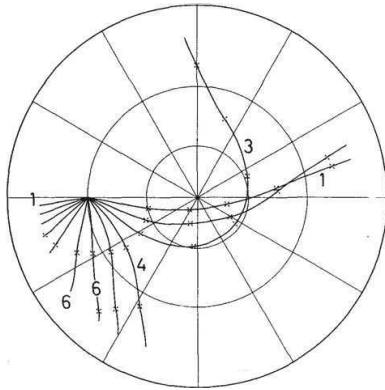
Link from the tropical sea-surface temperature anomalies to the mid-latitudes



RAYs

A theoretical calculation of propagation

Source at 15°, differencing background states.



A source in the tropics can have influence worldwide!

Hoskins and Karoly 1981.

PRIMITIVE EQUATIONS OF MOTION

Horizontal momentum:

$$\frac{Du}{Dt} - f v = -\frac{1}{\rho} \frac{\partial p}{\partial x}, \quad \frac{Dv}{Dt} + f u = -\frac{1}{\rho} \frac{\partial p}{\partial y}.$$

Hydrostatic:

$$\frac{\partial p}{\partial z} = -\rho g$$

Mass Conservation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

Thermodynamic:

$$\frac{D\theta}{Dt} = Q$$

Equation of state

$$p = \rho R T.$$

On the beta-plane: $f = f_0 + \beta y$.

VORTICITY AND POTENTIAL VORTICITY

Vorticity Equation:

$$\frac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega}_a \cdot \nabla)\mathbf{v} - \boldsymbol{\omega}_a \nabla \cdot \mathbf{v} + \frac{1}{\rho^2}(\nabla\rho \times \nabla\rho) \quad (1)$$

where $\boldsymbol{\omega}_a = \boldsymbol{\omega} + \mathbf{f}$.

Potential Vorticity:

$$\frac{DQ}{Dt} \equiv \frac{D}{Dt} \left(\frac{\boldsymbol{\omega}_a \cdot \nabla\theta}{\rho} \right) = 0. \quad (2)$$

Vertical component vorticity equation (approximate):

$$\frac{D}{Dt}(\zeta + f) = -(\zeta + f) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \left(\frac{\partial u}{\partial z} \frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \frac{\partial w}{\partial x} \right) \quad (3)$$

Three main assumptions:

1. Small Rossby number: $\frac{U}{fL} \ll 1$.
2. Scale similar to Rossby deformation scale: $L \sim \frac{NH}{f}$.
3. Variations in Coriolis parameter are small (cf. beta plane).

Potential Vorticity Equation becomes:

$$\frac{Dq}{Dt} = 0, \quad q = \zeta + f + \frac{\partial}{\partial z} \left(\frac{f_0 b}{N^2} \right).$$

$$u = -\frac{\partial \psi}{\partial x}, \quad v = \frac{\partial \psi}{\partial x}, \quad \zeta = \nabla^2 \psi + \beta y \quad b = f_0 \frac{\partial \psi}{\partial z}.$$

Perhaps the most important equation in dynamical meteorology!

BAROTROPIC (ONE-LAYER) QG EQUATION

Much of Rossby-wave and teleconnection theory is based on the single-layer (no vertical variation) equations:

3D equation:

$$\frac{Dq}{Dt} = 0, \quad q = \zeta + f + \frac{\partial}{\partial z} \left(\frac{f_0 b}{N^2} \right). \quad (4)$$

becomes

$$\frac{Dq}{Dt} = 0, \quad q = \zeta + f = \nabla^2 \psi + \beta y.$$

$$u = -\frac{\partial \psi}{\partial x}, \quad v = \frac{\partial \psi}{\partial x}, \quad \zeta = \nabla^2 \psi + \beta y \quad (5)$$

Begin with:

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla \zeta + \beta v = 0.$$

Linearize about a mean flow.

$$\psi = \Psi(y) + \psi'(x, y, t),$$

where $\Psi = -Uy$. Gives:

$$\frac{\partial}{\partial t} \nabla^2 \psi' + U \frac{\partial \nabla^2 \psi'}{\partial x} + \beta \frac{\partial \psi'}{\partial x} = 0.$$

Seek solutions

$$\psi' = \text{Re } \tilde{\psi} e^{i(kx + ly - \omega t)},$$

and obtain:

$$[(-\omega + Uk)(-(k^2 + l^2) + \beta k)] \tilde{\psi} = 0,$$

Gives *dispersion relation*:

$$\omega = Uk - \frac{\beta k}{k^2 + l^2}, \quad (6)$$

GROUP VELOCITY AND PHASE SPEED

Phase speed:

$$c_p^x = \frac{\omega}{k} = U - \frac{\beta}{k^2 + l^2}.$$

Group velocity:

$$c_g^x = U + \frac{\beta(k^2 - l^2)}{(k^2 + l^2)^2}, \quad c_g^y = \frac{2\beta k l}{(k^2 + l^2)^2}.$$

The wave activity (and sometimes the energy) in a Rossby wave propagates at the group speed:

$$\frac{\partial \mathcal{A}}{\partial t} + \nabla \cdot \mathbf{c}_g \mathcal{A} = 0. \quad (7)$$

This is one way teleconnections are made!

ROSSBY WAVES IN A VARYING MEDIUM

The Earth is not homogeneous! Even without topography.

Rossby wave equation:

$$\left(\frac{\partial}{\partial t} + U(y) \frac{\partial}{\partial x} \right) \zeta + v \beta^* = 0, \quad (\beta^* = \beta - U_{yy}) \quad (8)$$

Dispersion relation:

$$\omega \equiv ck = U(y)k - \frac{\beta^* k}{k^2 + l^2}, \quad (9)$$

Seek solutions:

$$\frac{\partial^2 \tilde{\psi}}{\partial y^2} + l^2(y) \tilde{\psi} = 0, \quad \text{where} \quad l^2(y) = \frac{\beta^*}{U - c} - k^2. \quad (10a,b)$$

For the experts, the WKB solution is:

$$\tilde{\psi}(y) = A_0 l^{-1/2} \exp \left(i \int l \, dy \right), \quad (11)$$

Group velocity:

$$c_g^x = U + \frac{(k^2 - l^2) \beta^*}{(k^2 + l^2)^2}, \quad c_g^y = \frac{2kl \beta^*}{(k^2 + l^2)^2}, \quad (12a,b)$$

Meridional Wavenumber:

$$l^2(y) = \frac{\beta^*}{U - c} - k^2.$$

But suppose $U = c$? Or $l = 0$?

Two regions will stop propagation:

1. A *turning surface*, where $l^2 = 0$ and $c_g^y = 0$,
2. A *critical surface*, where $U = c$ and l^2 becomes infinite, and $c_g^y = 0$.

Condition for wave propagation is that:

$$0 < U - c < \frac{\beta^*}{k^2}.$$

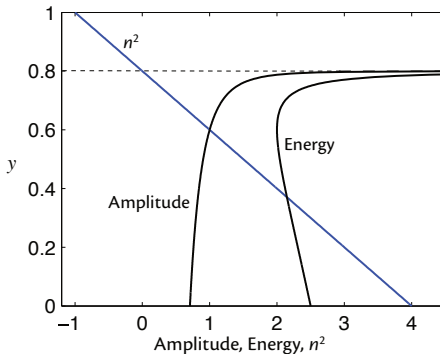
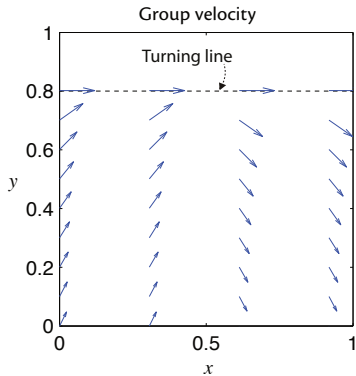
cf. Charney–Drazin condition.

A TURNING LINE

A turning line, where $l^2 = 0$. Group velocity becomes zonal.

$$c_g^x - U = \beta^*/k^2 \neq 0, \quad c_g^y \rightarrow 0.$$

Group velocity is purely zonal meridional wavenumber is large. The wave will *turn*.



$$(n = l)$$

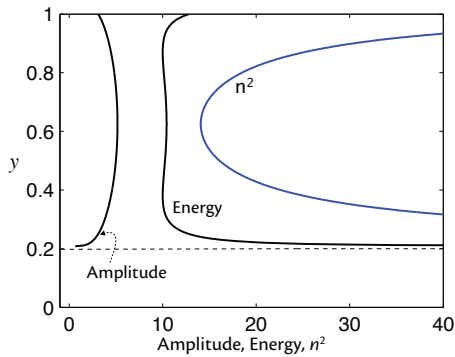
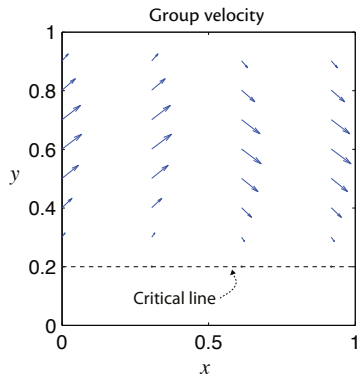
Amplitude ($\propto l^{-1/2}$) is infinite (in WKB calculation).

A CRITICAL LINE

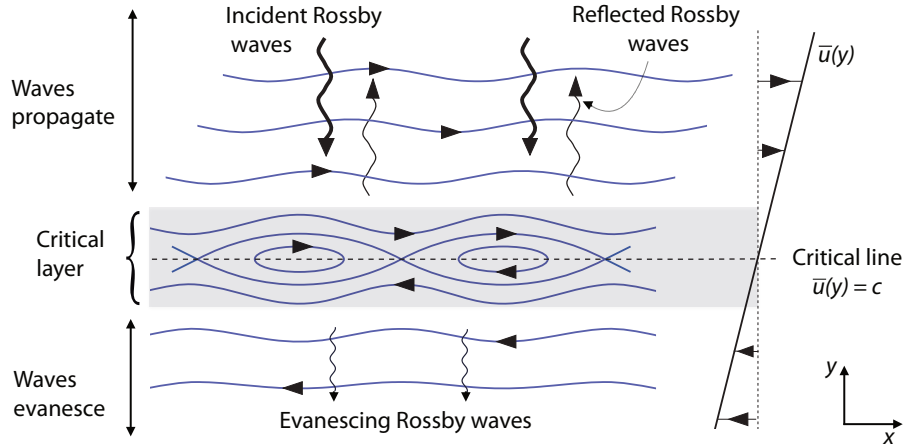
A critical line, where $c = U$ and l is large. The wave stops!

$$c_g^x - U \rightarrow 0, \quad c_g^y \rightarrow 0, \quad \frac{c_g^x - \bar{u}}{c_g^y} \rightarrow -\frac{l}{k} \rightarrow -\infty.$$

Group velocity in both directions is zero. The wave will break!



In fact WKB theory breaks down and we get critical layers of finite thickness.



- Ray theory tells us where waves go in inhomogeneous media, like the atmosphere.
- Similar approximations to those in WKB theory. Main one is:
 - Medium varies slowly compared to wavelength of wave.
 - Not well satisfied in atmosphere but proceed anyway!

Assume a solution of the form

$$\psi(\mathbf{x}, t) = a(\mathbf{x}, t)e^{i\theta(\mathbf{x}, t)} \approx a(\mathbf{x}, t)e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}.$$

with dispersion relation of the form

$$\omega = \Omega(\mathbf{k}; \mathbf{x}, t),$$

The wavenumber and frequency are functions of space and (in general) time.

After a bit of theory we find:

$$\frac{\partial \mathbf{k}}{\partial t} + \mathbf{c}_g \cdot \nabla \mathbf{k} = -\nabla \Omega,$$

and

$$\frac{\partial \omega}{\partial t} + \mathbf{c}_g \cdot \nabla \omega = \frac{\partial \Omega}{\partial t} = 0.$$

So frequency is constant along a ray.

We follow a wave packet using the ray equations:

$$\frac{d\mathbf{k}}{dt} = -\nabla\Omega, \quad \frac{d\omega}{dt} = 0, \quad \frac{d\mathbf{x}}{dt} = \mathbf{c}_g. \quad (1)$$

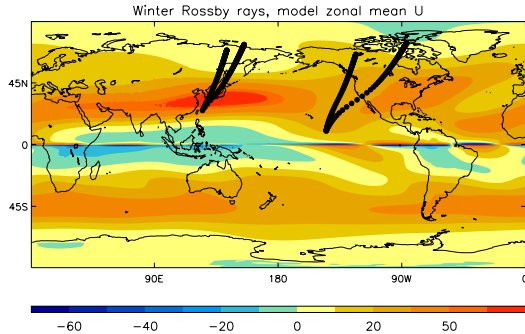
Recipe

- (1) Start with the location of the disturbance, and choose a wavenumber of interest (e.g. stationary wave, with $\omega = 0$).
- (2) Calculate the group velocity.
- (3) Move a short distance along in the group-velocity direction.
- (4) Re-calculate the x and y wavenumbers using (1a). The frequency stays the same.
- (5) Recalculate the group velocity.
- (6) Go to item (3).

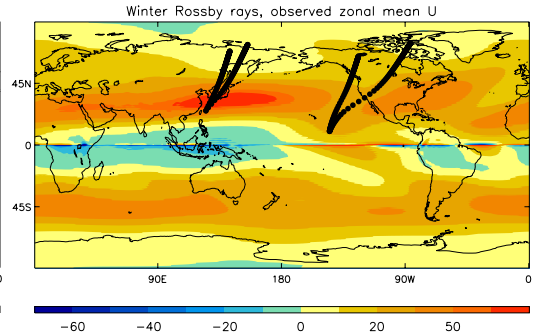
EXAMPLES

Use zonally-averaged zonal wind

Model



Observations

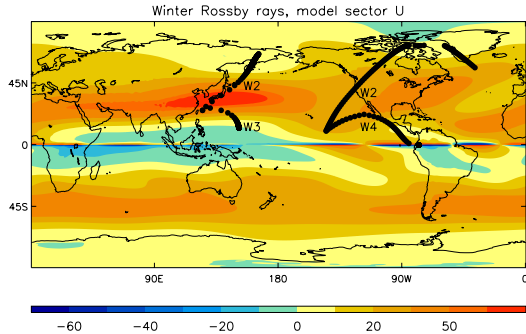


A. Scaife, UK Met Office

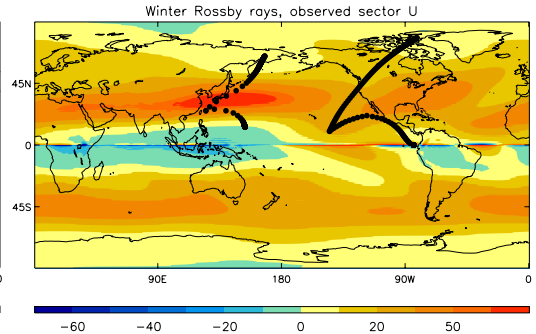
EXAMPLES

Use sector-averaged zonal wind

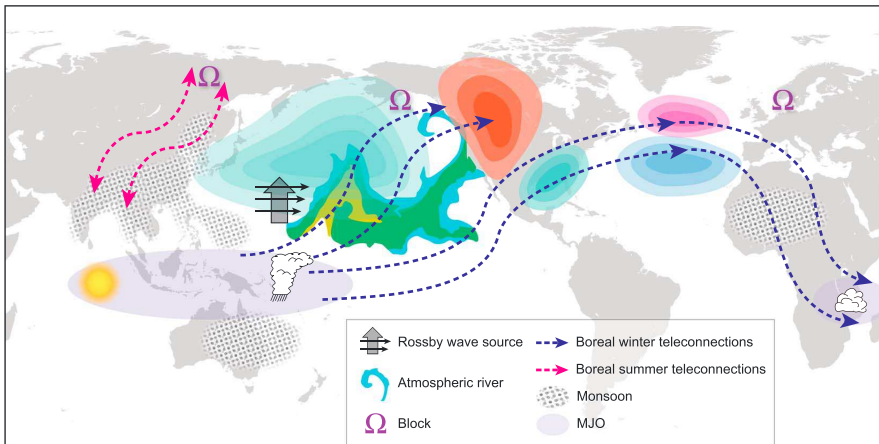
Model



Observations



A. Scaife, UK Met Office



Tropical–Northern Hemisphere interactions and teleconnections inferred from observational and numerical studies.

THEORY OF STATIONARY WAVES

- Stationary (Rossby) waves are a major factor in the large-scale zonally-asymmetric circulation.
- Due to orographic and thermal features at the surface of the planet.
- Understand using QG theory.

One-layer, orographic case

$$\frac{Dq}{Dt} = 0, \quad q = \zeta + \beta y - \frac{f_0}{H}(\eta - h_b), \quad (13)$$

Linear version:

$$\frac{\partial q'}{\partial t} + U \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0, \quad \text{or} \quad \frac{\partial}{\partial t} \left(\zeta' - \frac{\psi'}{L_d^2} \right) + U \frac{\partial \zeta'}{\partial x} + \beta v' = -U \frac{\partial \hat{h}}{\partial x}, \quad (14)$$

where $\hat{h} = h_b f_0 / H = h_b g / (L_d^2 f_0)$.

Solutions of the form:

$$(\zeta', \psi', \hat{h}) = \text{Re}(\tilde{\zeta}, \tilde{\psi}, \tilde{h}_b) \sin ly e^{ikx - \omega t} \quad \text{with} \quad \omega = \frac{k(UK^2 - \beta)}{K^2 + k_d^2}, \quad (15)$$

STATIONARY WAVES

Stationary solutions when $K = \sqrt{\beta/U}$ and

$$U \frac{\partial \zeta'}{\partial x} + \beta v' = -U \frac{\partial \hat{h}}{\partial x}, \quad (16)$$

giving

$$\tilde{\psi} = \text{Re} \frac{\tilde{h}_b}{(K^2 - K_s^2)}. \quad (17)$$

or, with friction

$$\tilde{\psi} = \text{Re} \frac{\tilde{h}_b}{(K^2 - K_s^2 - iR)}, \quad (18)$$

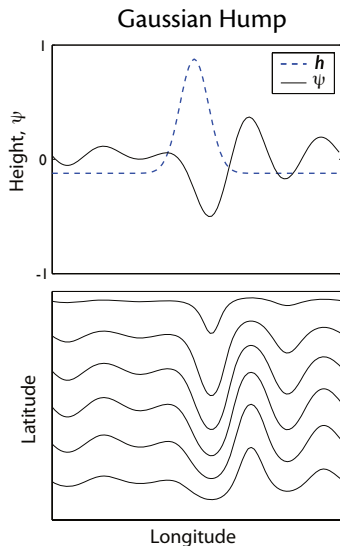
Important Points:

1. There is a resonant response, made finite by friction.
2. Solution can be in various phases with respect to the topography depending on whether $K > K_s$, $K < K_s$ or $K = K_s$ (resonant).
3. Typical for a mountain range to find a minimum in the streamfunction downstream of the ridge.

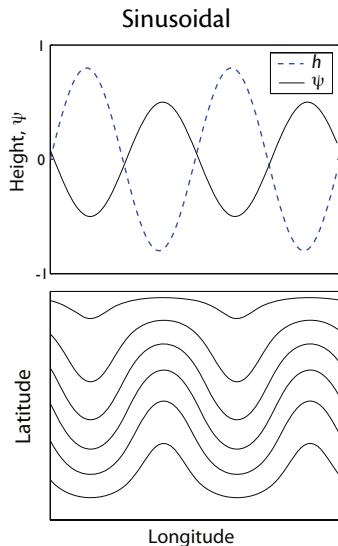
Some Solutions

Topography and
perturbation
streamfunction

Streamfunction



Resonant response, a 'trough', just downstream of the ridge.



No resonance. Response out of phase with the topography.

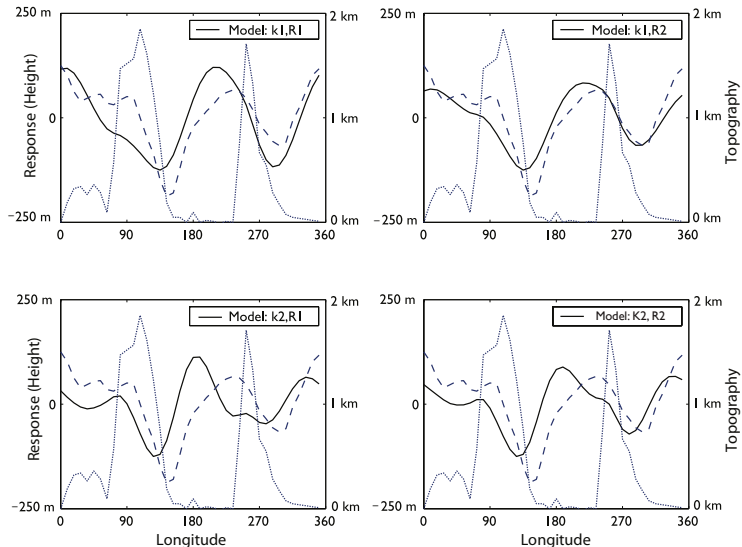
SOLUTIONS WITH EARTH TOPOGRAPHY

Different Frictional Coefficients and Zonal Winds

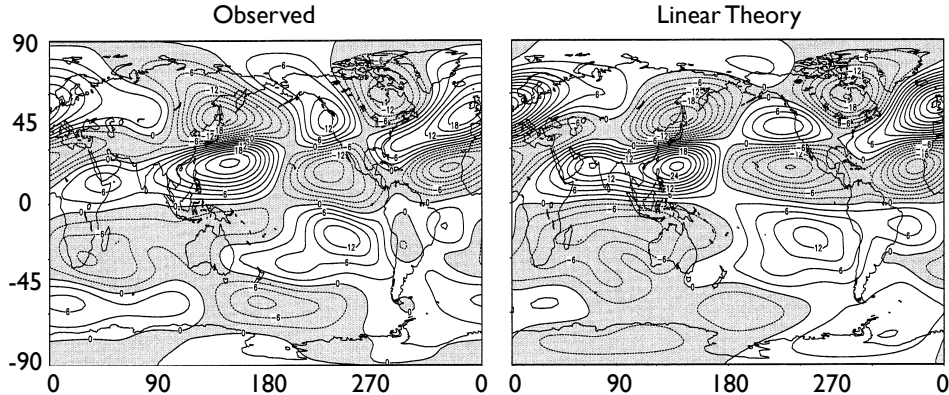
Dotted line:
Topography.

Dashed line:
solution.

Solid line:
observations



Time averaged streamfunction at 300 hPa.



Comprehensive linear GCM with the observed height-varying, zonally and time-averaged zonal winds.

Note much weaker zonal asymmetries in the southern hemisphere.

INGREDIENTS FOR TELECONNECTIONS

The Story so Far

1. Stationary Rossby waves: Determine background state.
2. Ray theory: tells us where waves from a disturbance go to.

Complications:

- Vertical propagation.
- Nonlinearity.
- Determining the wave source. Thermal effects.

Next Up

- (i) Numerical experiments on teleconnections
- (ii) The NAO.

TELECONNECTIONS AND OSCILLATIONS

EXPERIMENTS WITH GCMs

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Two TOPICS

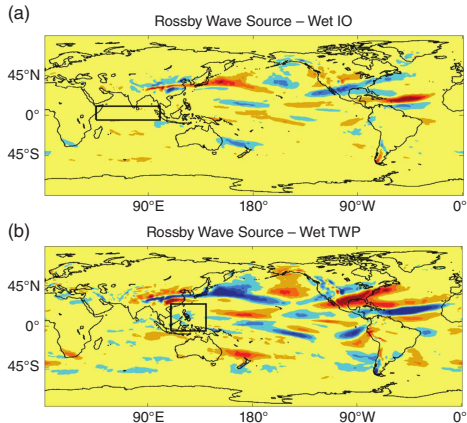


1. Teleconnections from Tropics to Mid-Latitudes
2. The North Atlantic Oscillation and Northern Annular Mode (NAO and NAM).

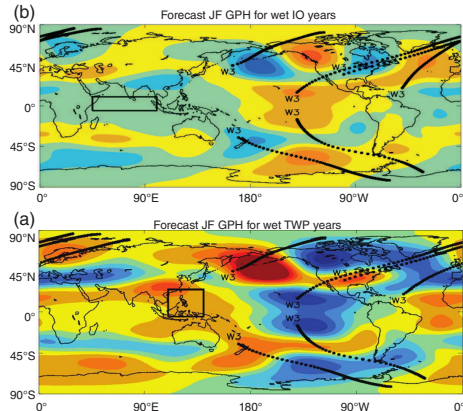
ROSSBY WAVE SOURCE AND WAVE TRAINS



Source: Vortex Stretching & Advection



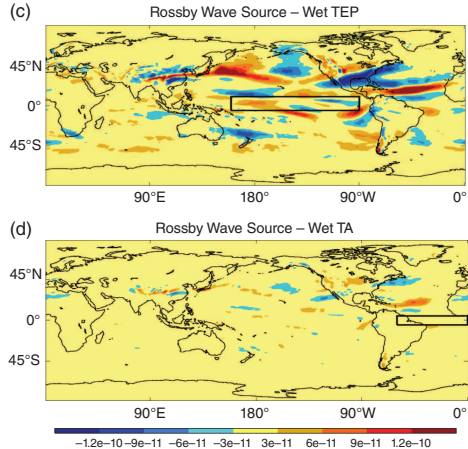
Wave Train



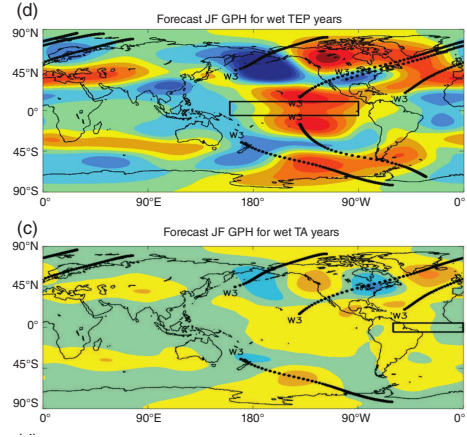
ROSSBY WAVE SOURCE AND WAVE TRAINS



Source: Vortex Stretching & Advection



Wave Train



EXPERIMENTS WITH AN IDEALIZED MODEL



Goals of the Study

How and why do SST anomalies affect Europe? In particular:

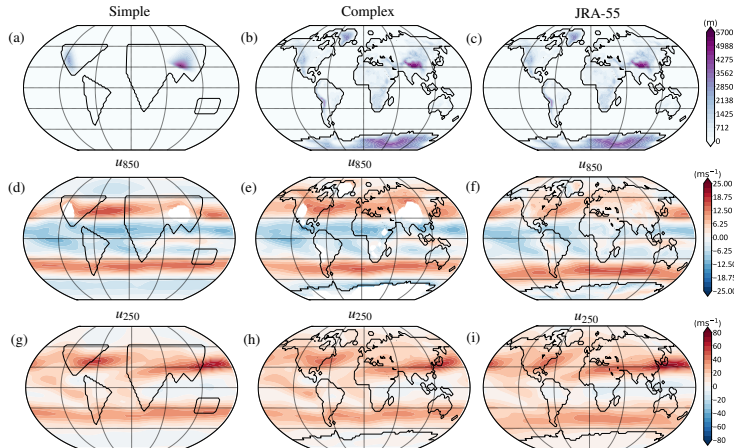
1. Tropical SST anomalies vs mid-latitude SST anomalies.
2. Winter vs summer.

It is generally thought that winter anomalies are more effective than summer ones.
And mid-latitude anomalies less effective than tropical ones. Why?

EXPERIMENTAL CONFIGURATION



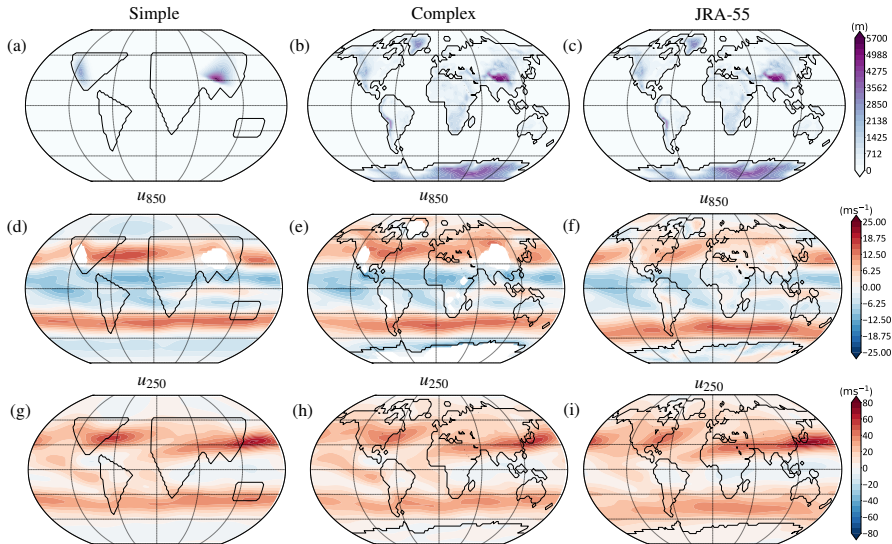
- (i) Impose very localized SST anomalies all over the global (mainly Pacific), in various seasons.
- (ii) Vary experimental configuration.
- (iii) Look for robust responses.



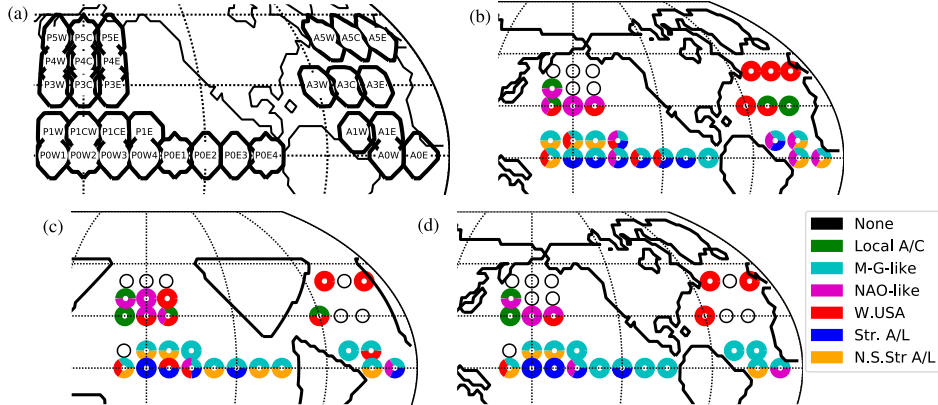
Winter

Note idealized and realistic continental outlines.

EXPERIMENTAL CONFIGURATION



LOCATION OF SST ANOMALIES



Each experiment is a single, persistent SST anomaly.

Look at summer and winter responses.



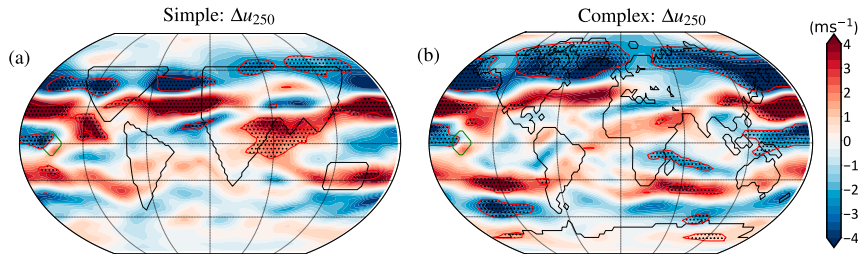
- Construct localized SST (7° wide) anomaly ($\sim 4^\circ \text{C}$) using Q fluxes (so that the SST anomaly arises from anomalous ocean heat transport). 31 locations, in two different configurations.
- Examine response and check for robustness with two conditions
 1. The response in a particular quantity (e.g. 250 hPa zonal wind) must be similar across the two configurations.
 2. The responses within each configuration must be statistically significant, as judged by the t-test with a 95% confidence limit.
 3. Look also at response to neighboring SST anomalies.



- **None**
- **Local anticyclonic circulation:** A statistically-significant anticyclone at 250 hPa, indicative of a local linear-like response as in Hoskins (1981).
- **Matsuno-Gill-like:** A statistically-significant response displaying the broad characteristics of the simple linear responses to tropical heating, specifically low-level eastward winds and upper-level westward winds.
- **NAO-like:** A statistically-significant response over the North Atlantic sector that looks like either a positive or negative NAO-like state. For some cases this will constitute a local response, and others (e.g. NAO-like responses to tropical-Pacific SST anomalies) it will constitute a remote teleconnection.
- **Anomaly off the Western US:** A statistically-significant small wind anomaly off the coast of Alaska.
- **Strengthened Aleutian-Low:** A statistically-significant strengthening of the Aleutian-Low in the central North-Pacific.

EXAMPLE RESPONSE

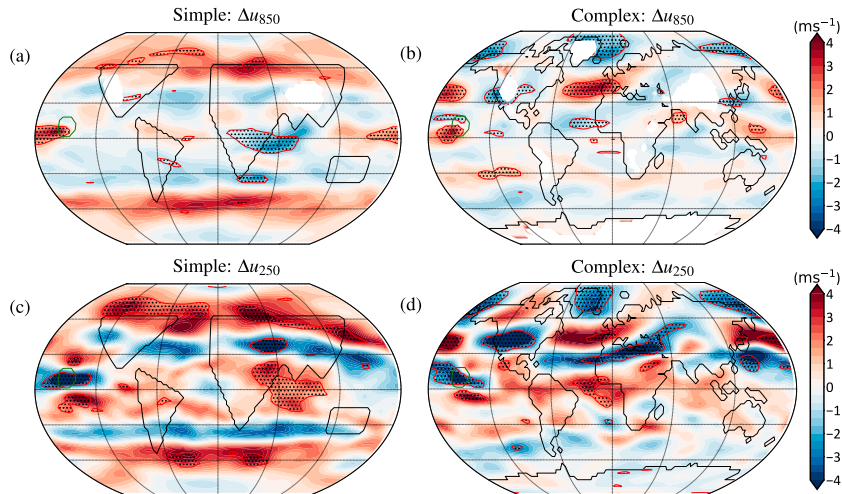
Western equatorial



Response of zonal wind in DJF to SST anomaly in western equatorial Pacific

EXAMPLE RESPONSE

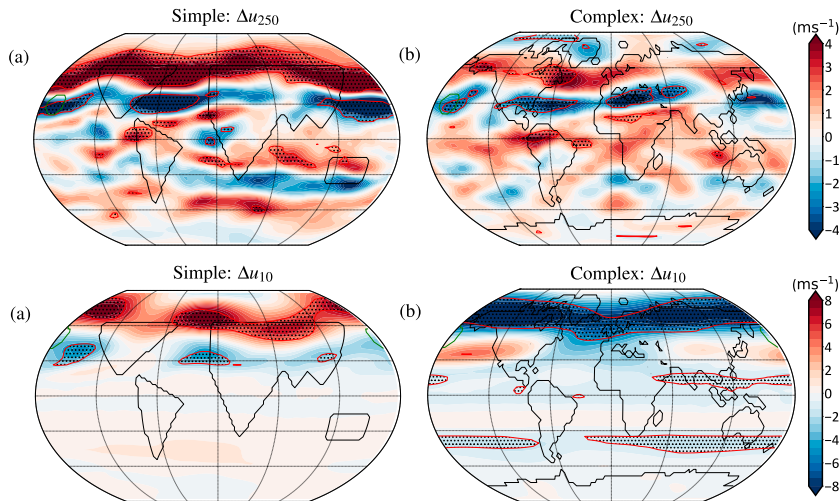
Central equatorial



Response of zonal wind in DJF to SST anomaly in central equatorial Pacific

EXAMPLE RESPONSE

Mid and high latitudes



SST anomaly in
mid-latitude Pacific

SST anomaly in high
latitude Pacific

RESULTS OF ANALYSIS



1. Tropical response is 'robust', especially SST anomalies in western and central Pacific. Mid-latitude response is weaker and less robust.
2. Winter Response is stronger than summer response.

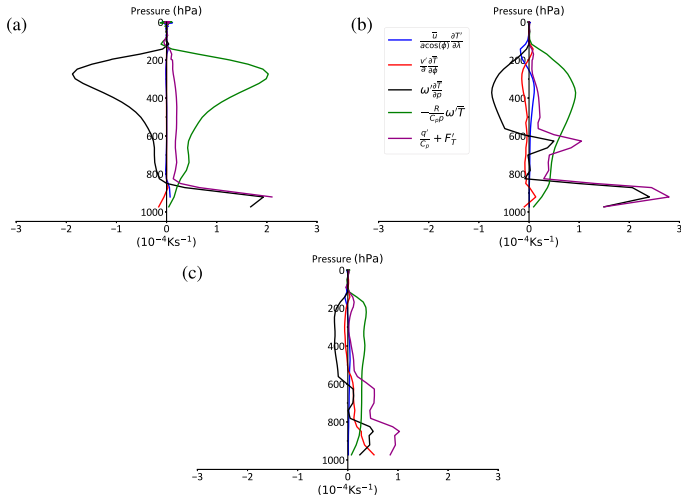
Why?

WHY THE TROPICS?

SST anomalies reach into the free atmosphere



Decomposition of temperature equation at three locations:



(a) = Central Equatorial Pacific, complex.

(b) = Central Pacific, 25° N, complex

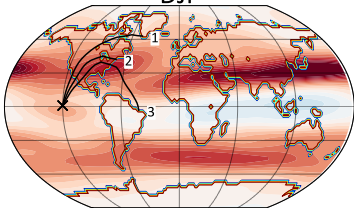
(c) = Central Pacific, 25° N, simple

Only (a) has a local response in the upper atmosphere.

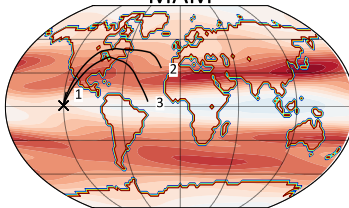
WHY WINTER NOT SUMMER?



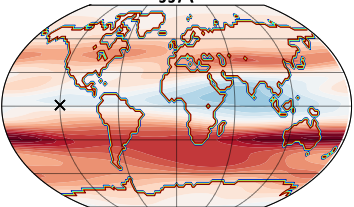
DJF



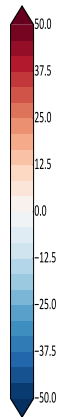
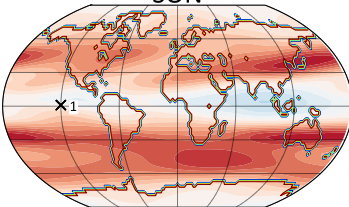
MAM



JJA



SON



Rays are trapped in the tropics in summer, but can escape in winter.

Wave propagation requires:

$$0 < U - c < \frac{\beta^*}{k^2}.$$

Need $U > 0$ for stationary waves.

Color shading is zonal wind

TELECONNECTIONS FROM THE TROPICS

Summary



Mid-latitudes vs Tropics

- (i) A tropical SST anomaly can reach the free atmosphere more easily than a mid-latitude anomaly.
- (ii) The response to a tropical anomaly is less sensitive to changes in the background climatology than a midlatitude anomaly.
- (iii) Mid-latitude SST anomalies exist in a highly variable environment with a low signal-to-noise ratio.

Summer vs Winter

- (i) A Rossby ray in winter can potentially propagate from the tropics to the mid-latitudes.
- (ii) A Rossby ray in summer is trapped in the tropics by westward winds:

$$l^2(y) = \frac{\beta^*}{U - c} - k^2.$$

So condition for wave propagation is that:

$$0 < U - c < \frac{\beta^*}{k^2}.$$

Need $U > 0$.

THE OTHER KIND OF TELECONNECTION

Examples: The NAO/NAM and ENSO



The NAO

It has been many times remarked, that the weather in Greenland is just the reverse to that in Europe; so that when the temperate climates are incommoded with a very hard winter, it is here uncommonly mild, and vice versa.

David Cranz, The History of Greenland

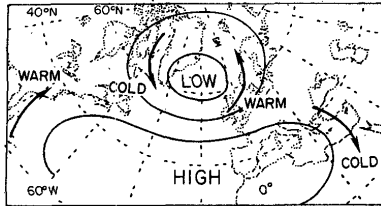


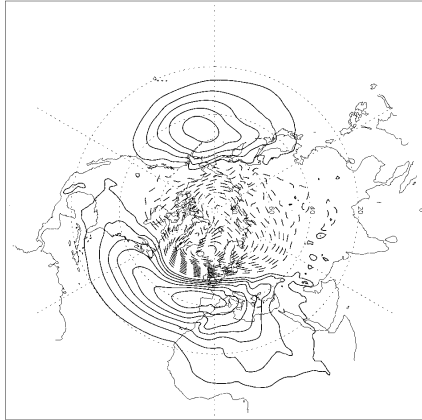
FIG. 1. Idealized relationships between pressure and temperature anomalies associated with the North Atlantic Oscillation.

Questions

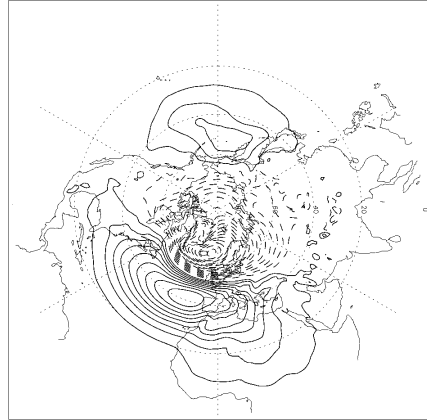
1. Why does it exist?
 - Variations of the storm track
2. Why does it change and on what timescales?

THE NAO

As an EOF



First EOF of hemispheric winter sea level pressure (25% of the variance).



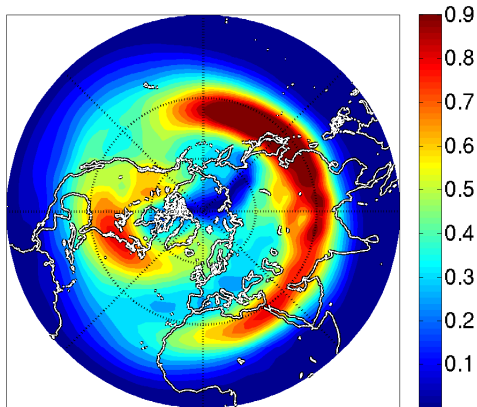
Right: first EOF of winter sea level pressure in the Euro-Atlantic sector only.
(Ambaum et al. 2001.)

NAO: EADY GROWTH RATE AND EDDY KE

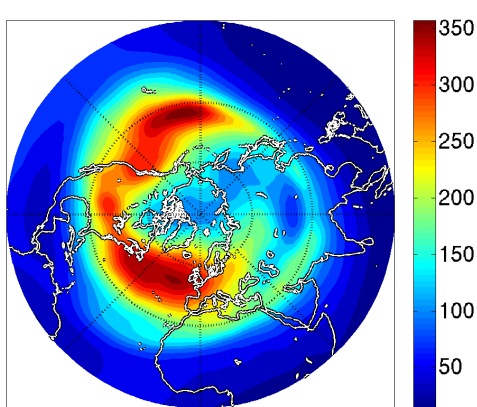
From Observations (re-analysis)



Eady Growth Rate



Eddy Kinetic Energy



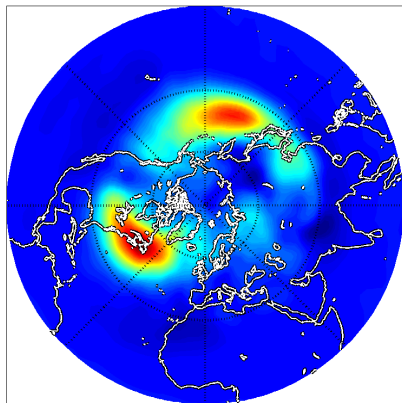
Eddy Kinetic Energy $\approx$ Storm Track Intensity $\approx$ co-incident with NAO

NAO: EDDY MOMENTUM AND HEAT FLUXES

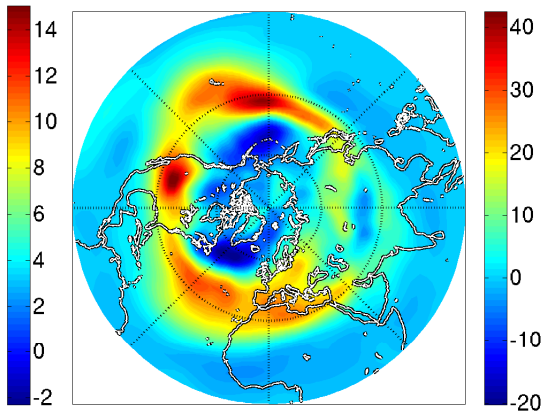
From Observations (re-analysis)



Eddy Heat Flux

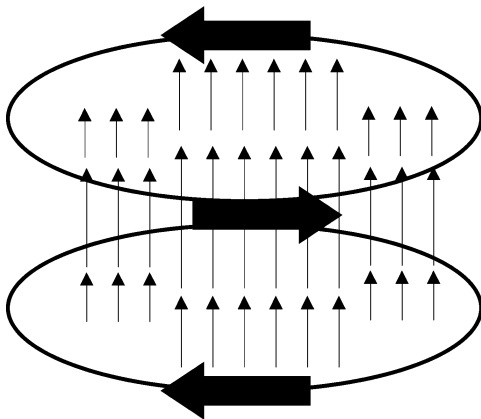


Eddy Momentum Flux (2–10 days)



Eddy momentum fluxes downstream of eddy heat fluxes.

MOMENTUM FLUXES AND EOFs

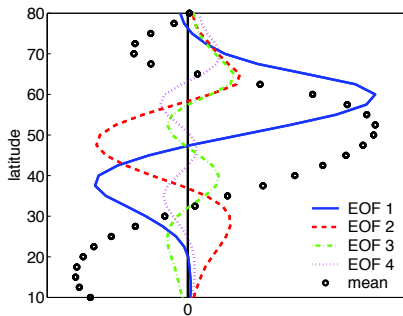


Anomalous momentum fluxes (arrows) give circulation pattern (contours) and hence EOFs.

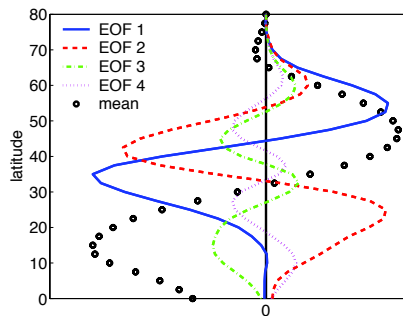
NAO: OBSERVED EOFs



Observed EOFs

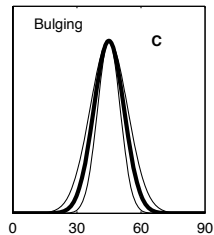
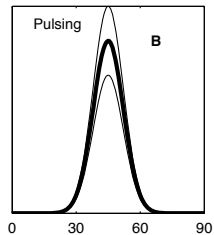
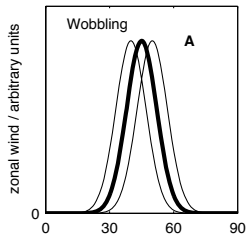


Zonally-averaged zonal wind EOF

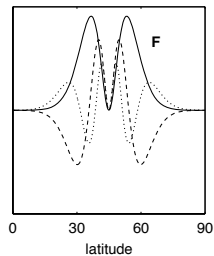
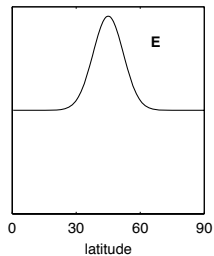
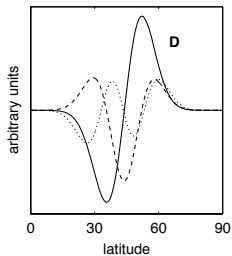


Angular momentum EOF.

NAO - JETS AND EOFs



Jet variability



Associated EOFs

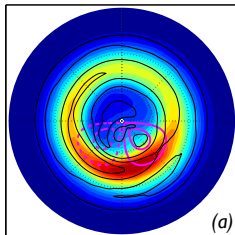
LOCALIZED EDDY FLUXES

The NAO and NAM

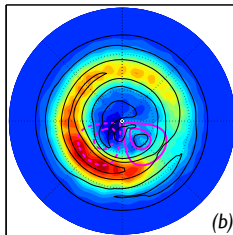
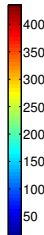


Add a mountain or a thermal anomaly and create a storm track.

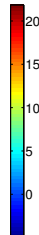
EKE



(a)

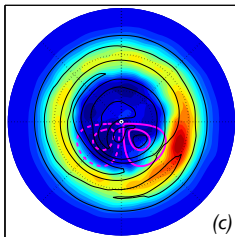


(b)

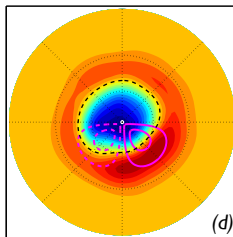
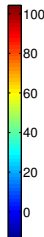


Eddy Heat Fluxes

Eddy Momentum
Fluxes



(c)

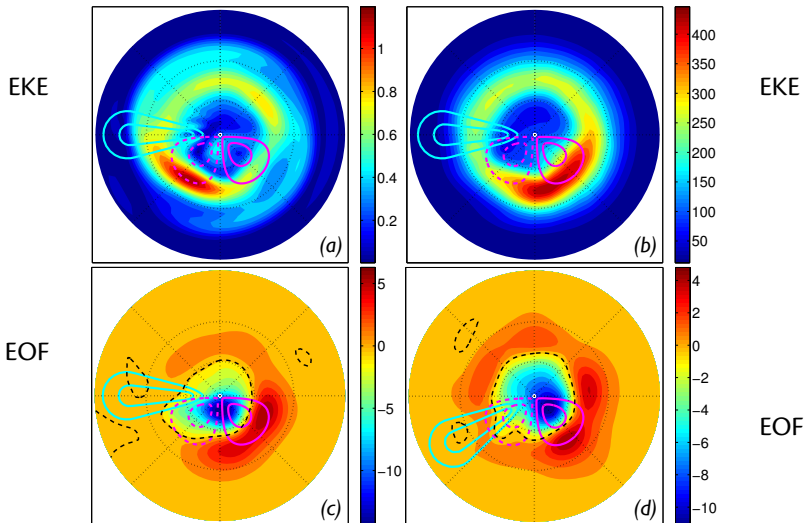


(d)



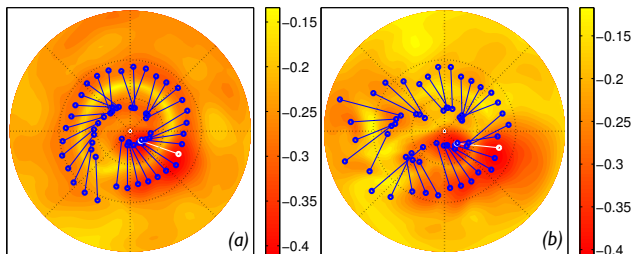
EOF

NAO - EDDY FLUXES AND EOFs

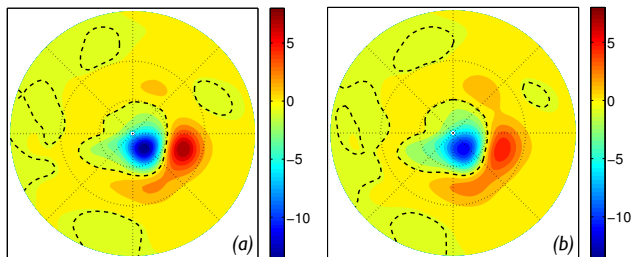




Teleconnection
Lines



Teleconnection
Patterns



Regress the pressure field onto the timeseries of pressure difference between the two most anticorrelated pairs.

$$T_c(t) = A \left[\frac{p'(\mathbf{x}_1, t)}{\sigma_{\mathbf{x}_1}} - \frac{p'(\mathbf{x}_2, t)}{\sigma_{\mathbf{x}_2}} \right]$$

$$P_c(\mathbf{x}) = \frac{1}{n} \sum_{t=0}^{t=n} p'(\mathbf{x}, t) T_c(t).$$

THE NAO TELECONNECTION

Some Conclusions



- The NAO is a see-saw type teleconnection
- Dipolar pattern – consequence of one baroclinic zone.
- Two phases: (i) Eddy momentum fluxes strong, mid-latitude jet and sub-tropical jet become distinct. (ii) Eddy momentum fluxes are weak subtropics, the mid-latitude jet is weak and the sub-tropical jet dominates.
- The NAO is the local manifestation of hemispheric dynamics (still a topic of debate).

OVERALL CONCLUSIONS



Teleconnections: Arise from a variety of sources and mechanisms.

1. Organized variability gives anti-correlations ('see-saws').
 - The NAO, NAM and SAM.
 - Storm track variability on synoptic timescales. SST variability can give year-to-year variability.
 - Stationary waves organize patterns into NAO.
 - Zonally-symmetric hemispheric-wide version is the SAM (and NAM).
 - ENSO (did not discuss that).
2. Long distance teleconnections, for example from tropics to mid-latitudes.
 - Rossby wave packets (ray tracing).
 - Stratospheric bridge.
3. Other mechanisms you will hear about this week!